SAFARI 3U: spectral bands, onboard model and expected product quality

Which Dragonfly Mantis mode SAFARI should fly (VNIR, SWIR or both), which bands, how large its onboard model must be, which Xiphos board can run it, and how accurate its fuel products are expected to be. From SAFARI's band-design study. 10 November 2026.

1. Introduction

SAFARI is the 3U wildfire-risk CubeSat described in the CUBICS 2026 application. Its payload turns each image into fuel-structure layers on board: fuel type, percent conifer, canopy closure and canopy height, the forest inputs of the Canadian Fire Behaviour Prediction System. This report gives the evidence behind the payload choices in the application's primary payload description: what each camera option can retrieve, what the onboard model needs, and what accuracy to expect. It is a companion to the EO coverage report (where and how often SAFARI images) and the RF antenna report (how the data comes down).

- **Section 2** describes the data and how every option was scored.
- **Section 3** compares the Mantis modes, VNIR against SWIR and both together.
- **Section 4** shows where the bands sit and why the standard filters suffice.
- **Section 5** sizes the onboard model and fits it to the Q7 and Q8 boards.
- **Section 6** gives the expected product accuracy and how it will be verified in orbit.
- **Section 7** sets out how far to trust the numbers; **section 8** lists the options and what comes next.

2. How the options were scored

A camera option is judged by what the onboard model can retrieve from it, not by band importance scores. For each option the model is retrained on exactly those bands and scored on ground it never saw. Every comparison is paired: two options are trained on the same splits and seeds, so their difference is measured directly.

Table 1. Data and method behind every result.

Element	What was used
Imagery, camera comparison	AVIRIS-NG airborne hyperspectral surface reflectance over boreal Canada and Alaska (NASA ABoVE), 5 nm sampling from 377 to 2496 nm, about 1.09 million development pixels. Each candidate band set is synthesised from the full spectrum with its filter response.
Imagery, band and model-size study	Sentinel-2 summer 2020 surface-reflectance composites over Canada's forest regions, 904 image tiles.
Labels	SCANFI v2 (Canadian Forest Service) 2020: percent conifer, canopy closure, canopy height, and a seven-class fuel family (conifer, deciduous, mixedwood, open, slash, non-fuel, water).
Model	A compact U-Net with four output heads, the same network that will run on board (about 31,000 parameters at the reference size).
Validation	Five outer folds of whole 1° blocks of ground, three training seeds each (15 fits per option). The test blocks are never seen in training or band selection.
Sensor noise	Every score includes a sensor noise model (placeholder values until Dragonfly publishes per-band SNR).
Score	Skill per target: 1 − RMSE ÷ RMSE of predicting the training mean (fuel family: balanced accuracy above chance). Combined skill is the mean of the four.

3. VNIR, SWIR or both: the Mantis modes compared

The Mantis is one 1U head with several modes: an RGB snapshot camera, a six-band multispectral (VNIR) pushbroom, a hyperspectral pushbroom over 470–900 nm, and a four-filter shortwave-infrared (SWIR) pushbroom over 1000–1750 nm, chosen from Dragonfly's standard filter catalogue. VNIR and SWIR share the fore-optic through a dichroic and can image together. Figure 1 and Table 2 compare every mode on the same ground.

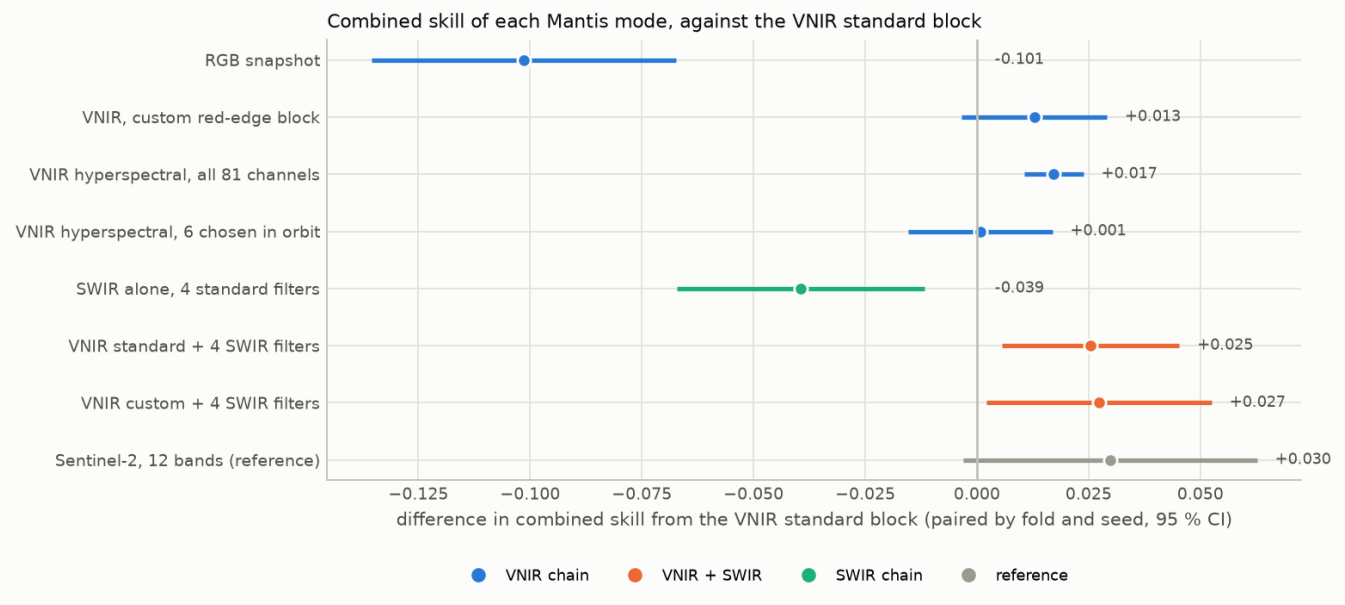


Figure 1. Combined skill of each Mantis mode minus the VNIR standard block, paired by fold and seed. Blue: VNIR chain only; orange: VNIR and SWIR together; green: SWIR only; grey: Sentinel-2 for reference.

Table 2. Each Mantis mode on AVIRIS-NG over boreal Canada and Alaska, with sensor noise. Highlighted: the three options compared head to head below.

Mantis configuration	Chain	Skill	Against VNIR standard [95 % CI]	Conifer RMSE (pts)	Closure RMSE (pts)	Height RMSE (m)	Family bal. acc.
RGB snapshot	VNIR	0.155	-0.101 [-0.136, -0.067]	27.1	19.1	4.25	0.35
VNIR, standard filter block	VNIR	0.256	–	24.3	16.4	3.97	0.47
VNIR, custom red-edge block	VNIR	0.269	+0.013 [-0.003, +0.029]	24.6	15.9	3.83	0.48
VNIR hyperspectral, all 81 channels	VNIR	0.273	+0.017 [+0.011, +0.024]	24.2	15.7	3.86	0.48
VNIR hyperspectral, 6 chosen in orbit	VNIR	0.257	+0.001 [-0.015, +0.017]	–	–	–	–
SWIR alone, 4 standard filters	SWIR	0.217	-0.039 [-0.067, -0.012]	26.2	17.4	4.06	0.44
VNIR standard + 4 SWIR filters	VNIR + SWIR	0.282	+0.025 [+0.006, +0.045]	24.2	15.4	3.79	0.49
VNIR custom + 4 SWIR filters	VNIR + SWIR	0.284	+0.027 [+0.002, +0.053]	24.2	15.2	3.82	0.49
Sentinel-2, 12 bands (reference)	–	0.286	+0.030 [-0.003, +0.063]	24.5	15.1	3.76	0.49

SWIR filters: Dragonfly L4 (1213 nm), L6 (1557), L9 (1599), L11 (1674), the catalogue filters nearest to the best SWIR placement found by searching the full spectrum. Hyperspectral 'chosen in orbit' means each fold selects its own 6 channels.

3.1 The SWIR camera against the VNIR camera

If SAFARI flies only one chain, VNIR is the better instrument: the standard VNIR block beats the four SWIR filters by +0.039 [+0.012, +0.067] in combined skill, and it is ahead on every product (Table 2): percent conifer 24.3 against 26.2 points, closure 16.4 against 17.4, height 3.97 against 4.06 m, fuel family 0.47 against 0.44. SWIR alone lacks the red and near-infrared contrast that separates conifer from broadleaf and canopy from open ground.

Flown together, the two are better than either. Adding the four SWIR filters to the standard VNIR block raises combined skill by +0.025 [+0.006, +0.045], about 10 %, to 0.282, close to Sentinel-2's 12 bands (0.286). Every product improves, closure most (16.4 to 15.4 points). With SWIR present, the custom red-edge VNIR block no longer matters (+0.002 [-0.004, +0.008]), so the standard block is enough.

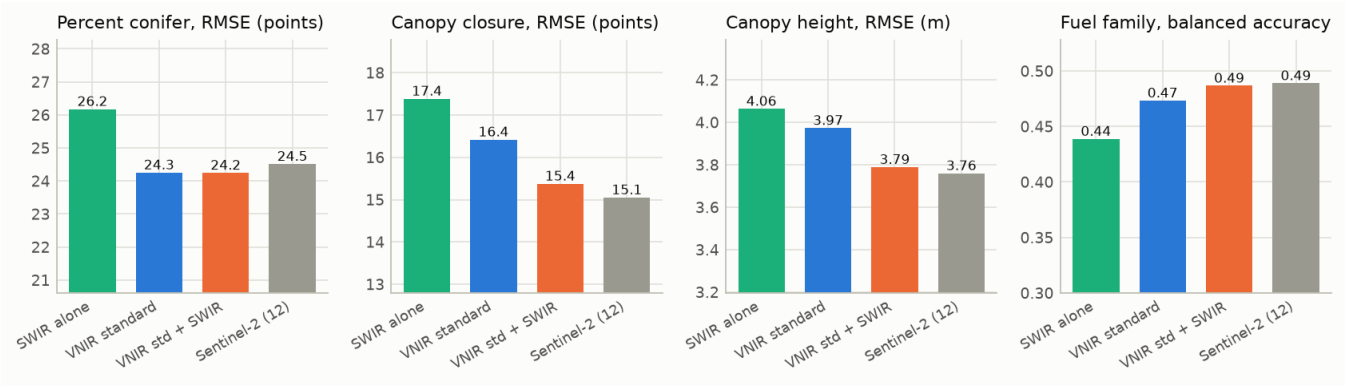


Figure 2. Per-product accuracy of SWIR alone, VNIR alone, both together, and Sentinel-2. Lower is better for the three RMSE panels, higher for balanced accuracy.

Table 3. The two Mantis chains and their combination as options.

	VNIR only (standard block)	SWIR only (4 filters)	VNIR + SWIR together
Combined skill	0.256	0.217	0.282
Bands	480, 545, 660, 780, 900 nm + panchromatic	1213, 1557, 1599, 1674 nm	all ten
Detector	silicon CMOS	standard InGaAs (top filter near its 1.7 µm cut-off)	both
GSD and swath at 500 km	32 m (16 m pan), 32 km	to confirm: 32 m and 32 km if it shares the head's optics, 14.8 m and 18 km on an earlier parts sheet	as each chain
Revisit (nadir)	≈40–100 days for most altitudes (EO report)	longer if the swath is 18 km: median ≈100 days at 65° N, ≈160 at 45° N	set by the narrower swath where both are needed
Readiness	TRL 9 (Dragonfly flight heritage)	TRL 8 on Dragonfly's parts sheet (to confirm)	TRL 8
Onboard inputs	11 channels (5 bands + 6 indices)	4 bands + indices	more channels; larger first layer
What it adds	the fire products and vegetation indices	moisture-sensitive bands; snow and cloud discrimination	+0.025 skill over VNIR
What it costs	baseline	second focal plane, InGaAs power and thermal control	both chains' power, mass and data

Power, mass and SWIR geometry for the combined head are to be confirmed with Dragonfly.

The SWIR filters need no custom order: Dragonfly's catalogue filters L4, L6, L9 and L11 score within 0.002 of the best SWIR placement found by searching the full spectrum (-0.002 [-0.005, +0.002]). Searched separately in each fold, the four filters lose most of their gain to the search's own noise (Table 2 note), so they must be fixed in advance.

3.2 Does a larger model change the ranking?

The comparison above uses one model size. To check that the ranking is not a property of that size, each camera's model was retrained at seven sizes, from 16 × 2 to 256 × 3 (hidden units × layers, 500 to 137,000 parameters), on the same folds and seeds. VNIR beats SWIR alone at every size, by +0.032 to +0.041; the larger networks do not let the SWIR camera catch up. Adding SWIR to VNIR gains +0.025 to +0.033 at every size. At the largest size the gain is +0.032 [-0.002, +0.066], so its interval just reaches zero there, while it clears zero at five of the seven sizes. Each camera's best size is small: VNIR 64 × 3, SWIR 128 × 2 and both together 256 × 2, each within 0.007 of the 64 × 2 size every band comparison used.

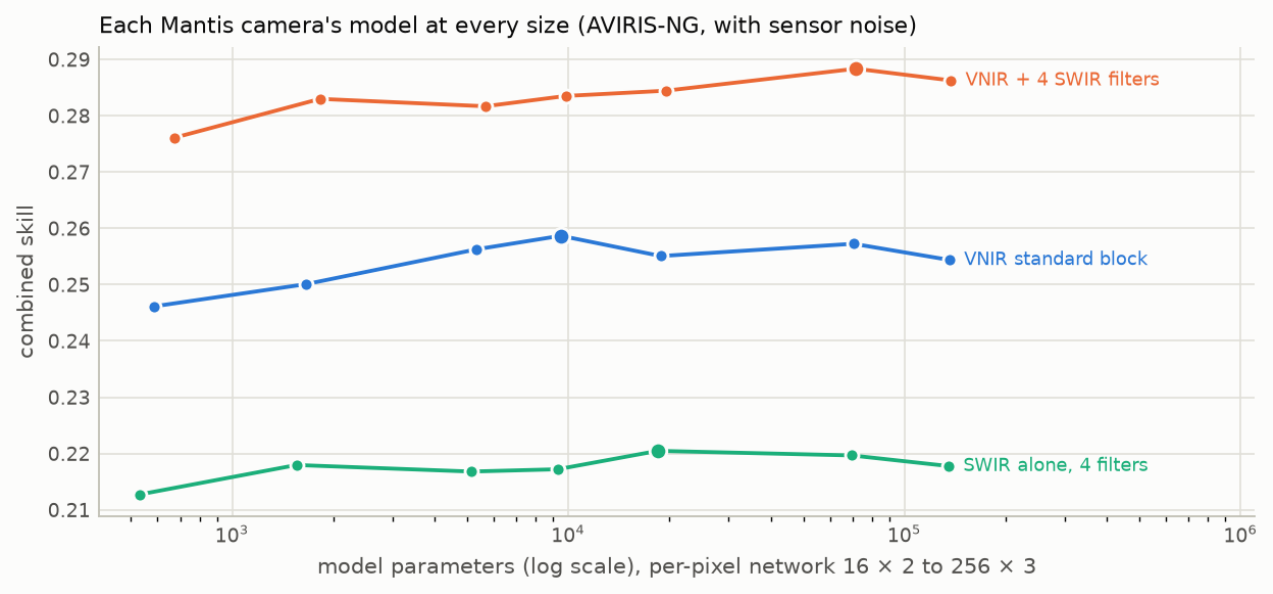


Figure 3. Combined skill of each Mantis camera's per-pixel model at seven sizes. Larger dot: that camera's best size. Paired by fold and seed (15 fits per point).

4. Where the bands sit

Searching Sentinel-2's bands in each training fold and keeping the bands most folds agree on gives two stable band sets. The five-band set B4, B5, B8A, B11 and B12 (red, red edge, narrow near-infrared and both SWIR bands) scores 0.405, 0.010 ± 0.003 below the best band set found (0.416). An eight-band set ties the best (0.416) but is less stable across folds.

A Sentinel-2 stand-in for the Mantis standard VNIR block (B2, B3, B4, B7, B8A; B7 matches the Mantis 780 nm band) comes within 0.003 ± 0.002 of the stable five-band set on the same splits. On Sentinel-2, VNIR alone gives up little; on the hyperspectral data (section 3) SWIR adds more, because it is measured with four narrow filters rather than two broad bands.

The custom red-edge VNIR block (490, 560, 665, 705, 740, 865 nm) beats the standard block by +0.013 [-0.003, +0.029]: not significant on its own, and nil once SWIR is added. The standard block is therefore flown, with no custom filter procurement.

5. The onboard model and the processing board

The onboard network was trained at twelve sizes (width 4–32 channels, depth 1–3) at the VNIR stand-in bands. Accuracy rises with width up to 32 channels and gains nothing from extra depth. The best size is 32 × 1 (107 k parameters, 65 GMAC per 1000 × 1000 px scene). The 8 × 2 size used for every band comparison is 0.0113 ± 0.0018 below it, and 16 × 1 is 0.0046 ± 0.0005 below.

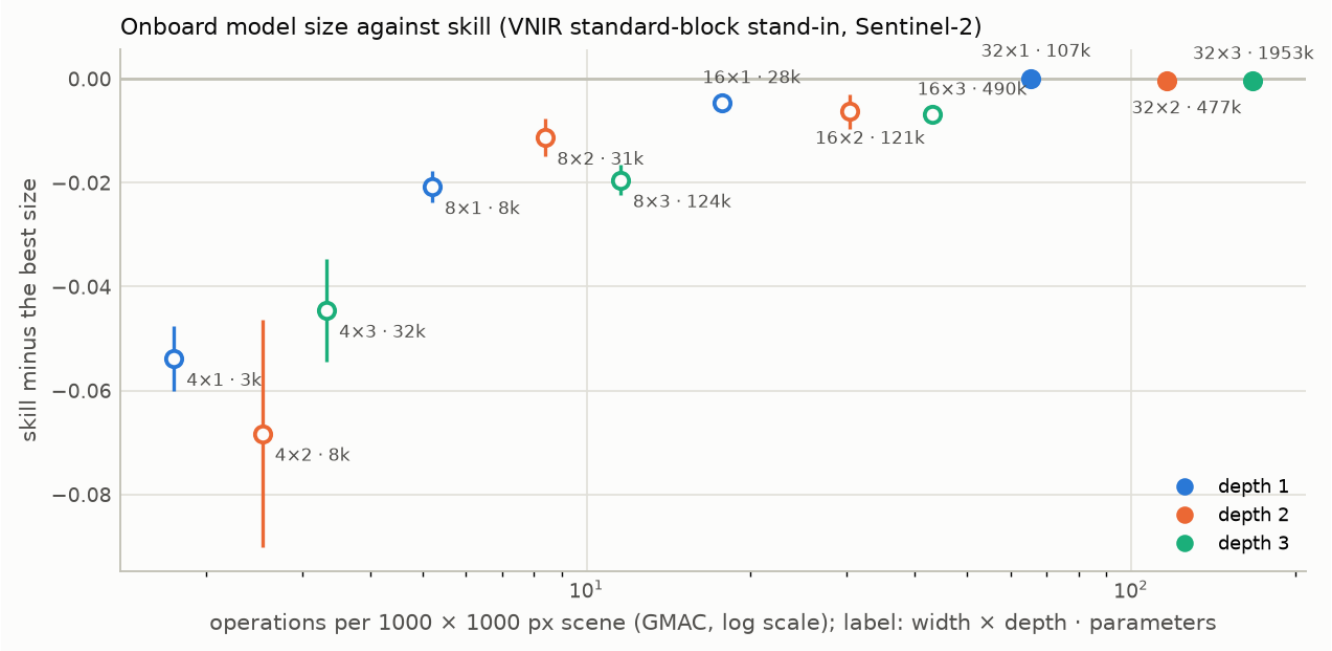


Figure 4. Combined skill of each model size minus the best, against the operations per scene. Filled: tied with the best within one paired standard error.

Table 4. Candidate model sizes and their processing time per scene on each board (budget 43 s).

Size	Params	GMAC / scene	Skill	Against best	Conifer / closure / height / family	Q8 AI engine	Q7 dataflow logic
8 × 2	31 k	8.4	0.402	−0.011	23.7 / 13.8 / 4.54 / 0.53	0.02 s	9 s
16 × 1	28 k	17.7	0.409	−0.005	23.5 / 13.6 / 4.49 / 0.53	0.03 s	35 s
32 × 1	107 k	65.3	0.413	best	23.2 / 13.5 / 4.44 / 0.53	0.11 s	138 s (over budget)

Board times are from converting the network through each vendor toolchain (Q8: Vitis AI engine, 8-bit; Q7: FINN dataflow logic, 4-bit, at minimal parallelism) with 22 inputs, so they bound the 11-input VNIR model from above. A wider FINN fold would speed the Q7 at the cost of more logic; its fit is not yet measured.

Table 5. The two processing boards (Xiphos datasheets).

	Xiphos Q7	Xiphos Q8
Processor	Zynq-7020, 2 × Cortex-A9 at 766 MHz	Zynq UltraScale+ ZU7EG, 4 × Cortex-A53, 2 × Cortex-R5
Memory	1 GB DDR3L (512 MB with EDAC)	4 GB LPDDR4 with EDAC, 2 × 128 GB eMMC
Power, mass	2 W typical, 24 g	5 W typical, 56 g
Flight heritage	Q7S in orbit since June 2016	Q8S in orbit since September 2020
Runs the best size (32 × 1)	Only with a wider dataflow fold (unmeasured)	Yes, ≈0.1 s
Runs 16 × 1	Yes, ≈35 s, about 19 % of its logic	Yes, ≈0.03 s

Model size and board are therefore one decision. The Q8 runs the best model with a large timing margin and real-time capability at 5 W. The Q7 runs a model 0.005 below the best within the budget at 2 W.

The same twelve sizes were then trained at three more band sets: the stand-in with B11 added (the one Sentinel-2 band inside the SWIR chain's range), and the two consensus sets of section 4. Width 32 is the best size at every band set, and 16 × 1 stays within 0.009 of it, so the size decision does not depend on which bands are flown. Adding B11 to the stand-in raises the best size's skill by +0.013 [+0.010, +0.016], so the SWIR chain's gain holds for the onboard U-Net as well as for the per-pixel models of section 3.

Table 6. The onboard U-Net size ladder at four band sets (15 fits per size).

Band set (Sentinel-2)	Best size	Skill	16 × 1 below best	8 × 2 below best
Mantis VNIR stand-in (B2 B3 B4 B7 B8A)	32 × 1	0.413	−0.005	−0.011
stand-in + B11 (one SWIR band)	32 × 2	0.427	−0.007	−0.021
stable five-band set (B4 B5 B8A B11 B12)	32 × 3	0.426	−0.008	−0.021
eight-band set (tied with the best)	32 × 1	0.438	−0.007	−0.022

Adding SWIR widens only the network's first layer, so the board times above change little; the FINN and DPU builds at these exact inputs are still to be run.

6. Expected product quality and its verification

Table 7. Expected accuracy of the onboard fuel layers (AVIRIS-NG, Mantis bands, with sensor noise, 8 × 2 model).

Product	VNIR only	VNIR + SWIR	Verified in orbit against
Percent conifer (RMSE, points)	24.3	24.2	national forest-inventory maps and plots
Canopy closure (RMSE, points)	16.4	15.4	national forest-inventory maps and plots
Canopy height (RMSE, m)	3.97	3.79	inventory maps; ICESat-2 canopy heights
Fuel family (balanced accuracy, 6 classes, chance 0.17)	0.47	0.49	inventory maps; the CWFIS fuel-type map

The Sentinel-2 size study (section 5) gives slightly better closure and slightly worse height on its own data; the larger model improves each by a few percent.

In orbit, onboard products will be checked against ground reprocessing of the same downlinked raw scenes, radiometry against near-simultaneous Sentinel-2 overpasses, and the wildfire-risk polygons against the burned areas and fire starts that follow, as set out in the application's measurable science objectives.

7. How far to trust these numbers

- **Solid.** The ranking of the Mantis modes: every comparison is paired over 15 fits on ground the model never saw, and the three anchor configurations rerun in a second job reproduced their scores exactly.
- **Depends on a measurement.** Absolute accuracy depends on Mantis's real signal-to-noise ratio (placeholder until Dragonfly publishes it) and on the SWIR chain's real geometry. The labels are themselves mapped from Landsat, so their own error is inside every RMSE.
- **Not yet modelled.** The 32 m GSD against 30 m labels; the panchromatic band in the model; INT8 or 4-bit quantisation on the board (to be measured on engineering models before PDR).

8. Options and next steps

Camera. (a) VNIR standard block only: TRL 9, simplest, the baseline in the application. (b) VNIR + four SWIR catalogue filters: +0.025 combined skill (about 10 %), near Sentinel-2's 12 bands, at the cost of the SWIR chain's power, mass and a possibly narrower swath. (c) SWIR only: not recommended on this evidence; 0.039 below VNIR alone, and behind it at every model size tried (section 3.2).

Model and board. (a) 32 × 1 on the Q8: best accuracy, real time, 5 W. (b) 16 × 1 on the Q7: 0.005 below the best, within budget, 2 W. (c) 8 × 2 on either: the reference size, 0.011 below the best.

- Done since the first draft: the size ladder at the true Mantis bands for VNIR, SWIR and both (section 3.2), and at four Sentinel-2 band sets (section 5). The rankings held.
- Convert the chosen size at the chosen bands through FINN and the DPU (the board times above used 22 inputs).

- Benchmark the chosen size on Q7 and Q8 engineering models: quantised accuracy, latency, power.
- Confirm with Dragonfly: per-band SNR, the SWIR chain's GSD, swath and TRL, and the combined head's power and mass.

Sources

- SAFARI Stage 1 band-design study (this repository): results packages P12/p12_sensors (Mantis modes, runs sensors_v1 and sensors_fixed), P12/p12_bands (Sentinel-2 and AVIRIS-NG band consensus), P12/p12_arch (onboard model size at four band sets), P12/p12_vnir_swir (VNIR against SWIR at every model size), P12/p12_convert (FPGA conversion), all run on the Digital Research Alliance of Canada clusters (Narval, Nibi), November 2026.
- NASA ABoVE airborne campaign, AVIRIS-NG surface reflectance over boreal Canada and Alaska (5 nm sampling, 377–2496 nm), convolved to each candidate band set.
- Sentinel-2 Level-2A surface reflectance, summer 2020 median composites (ESA Copernicus).
- SCANFI v2 (Canadian Forest Service), 30 m maps of species crown closure, canopy closure and height for 2020, used as the training and test labels.
- Dragonfly Aerospace, Mantis datasheet, band tables and SWIR filter catalogue (6 November 2026).
- Xiphos, Q7 specification sheet and Q8 specification sheet XTI-2001-2024-i Rev C (13 August 2026).
- SAFARI 3U: Earth observation coverage, resolution and revisit (EO coverage report), and SAFARI 3U: the lowest-cost antennas for guaranteed contact (RF antenna report).
- SAFARI CUBICS 2026 application (draft): mission, objectives, products and schedule.